

# VIDEO VIOLENCE DETECTION

A Machine-Learning System for Detecting Violence in Short Videos

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## PROJECT GOAL & PROBLEM

Public spaces are widely covered by surveillance cameras, but continuous manual review is difficult, leading to missed critical events and delayed response times.

### OBJECTIVE

Build an intelligent system to classify short videos into:

- Violence (1):**  
Aggressive physical actions.
- Non-Violence (0):**  
Normal routine activity.

## DATASET & PREPROCESSING

- Source:** XD-Violence dataset.
- Data split** into Train, Validation, and Test sets.
- Diverse clips** with violence and non-violence, varying quality, lighting, and camera angles.
- Video Slicing:** Uniform 3-second clips.
- Frame Extraction:** 16 frames per clip.
- Resize & Normalization:** Frames resized to 224×224 and normalized.
- Visual-only analysis:** classification is based on video frames, not audio.

## SOLUTION ARCHITECTURE

The final system uses MViT V2 S (Multiscale Vision Transformer) to capture spatial and temporal patterns in short video clips.

- Spatial Modeling:** learns objects and context within frames using self-attention.
- Temporal Modeling:** captures motion patterns across consecutive frames.
- Training:** transfer learning and fine-tuning for binary violence classification.
- Optimization:** BCE loss and AdamW.
- Regularization:** Dropout, weight decay, and early stopping.

### PROJECT TECHNOLOGY INFRASTRUCTURE

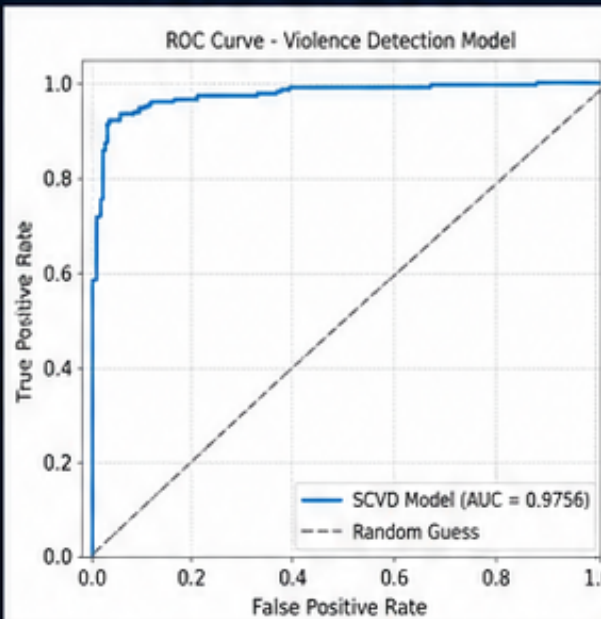


## EVALUATION & RESULTS

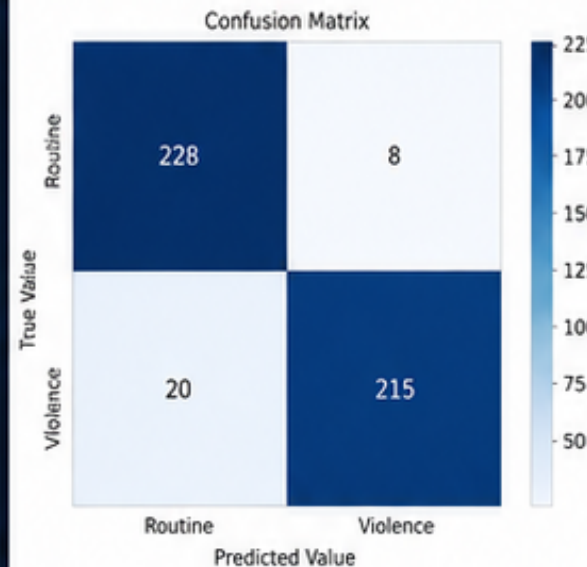
The initial baseline (ResNet50 + LSTM) served as a reference model, but the final MViT V2 S model achieved a robust improvement.

| METRIC              | BASELINE | FINAL MODEL | NOTES                   |
|---------------------|----------|-------------|-------------------------|
| Validation Accuracy | 52.6%    | 96%–97%     | Strong improvement      |
| F1-Score            | 0.674    | ~0.96       | Better balance          |
| Precision           | 0.513    | High        | Fewer false alarms      |
| Recall              | 0.981    | High        | Strong detection        |
| ROC-AUC             | —        | 0.9756      | Strong class separation |

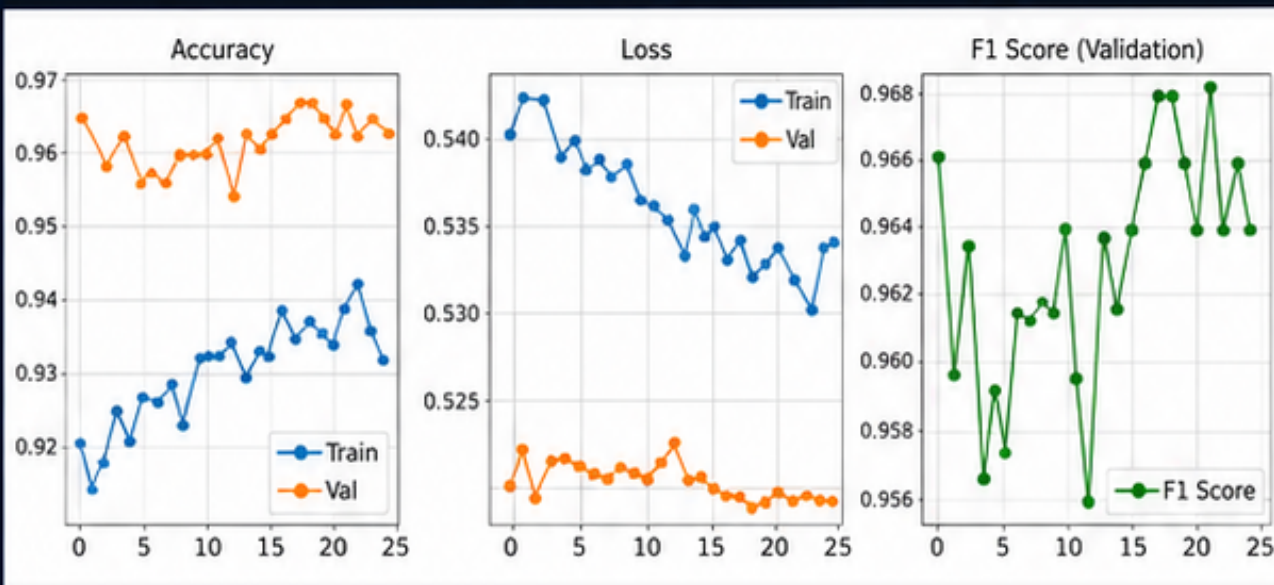
### ROC CURVE (AUC = 0.9756)



### FINAL CONFUSION MATRIX



## TRAINING PROGRESS (ACCURACY, LOSS, F1-SCORE)



## PROJECT VALUE & IMPACT

- Automated Screening:** Supports scalable review of large volumes of surveillance video.
- Resource Optimization:** Reduces manual monitoring load and reviewer fatigue.
- Faster Response Potential:** Can support faster identification of suspicious video segments in future monitoring systems.

## FUTURE HORIZONS

- Real-time Stream Inference:** Future integration with live video streams.
- Spatio-Temporal Localization:** Localizing violent incidents within the scene.
- Multi-Class Expansion:** Distinguishing between different incident types.



This project demonstrates a strong step toward intelligent video understanding for public safety. The MViT V2 S model achieves high accuracy and robustness, enabling reliable detection of violence in short video clips under diverse real-world conditions.

